

In-Class Assignment: DDPG Reinforcement Learning

Please answer the following questions.

1. What type of environment is DDPG primarily designed for?

- A) Environments with discrete action spaces (Ex, up, down, left, right)
- B) Environments with continuous action spaces (Ex, steering wheel angles, joint torque)
- C) Environments with no rewards
- D) Environments where the state space is completely known beforehand

2. Which reinforcement learning architecture does DDPG use?

- A) Monte Carlo Tree Search
- B) Value Iteration
- C) Actor-Critic
- D) REINFORCE

3. What does the "Deterministic" in DDPG mean regarding its policy?

- A) The agent outputs a specific, exact action value for a given state rather than a probability distribution.
- B) The agent always takes a random action.
- C) The agent perfectly predicts the environment's next state.
- D) The agent ignores the current state and repeats its previous action.

4. How does an agent using DDPG explore its environment, given that its policy is deterministic?

- A) By using the ϵ -greedy strategy (taking random actions a percentage of the time).
- B) By widening the variance of its action probability distribution.
- C) By randomly resetting the environment's state.
- D) By adding noise (such as Gaussian or Ornstein-Uhlenbeck noise) directly to the action output.

5. DDPG is an off-policy algorithm. Which two mechanisms does it borrow from DQN (Deep Q-Network) to stabilize learning?

- A) Trust Regions and Gradient Clipping
- B) Replay Buffer and Target Networks
- C) Policy Iteration and Value Iteration
- D) Monte Carlo rollouts and Bootstrapping